

# Samim Reza

Dhaka, Bangladesh

+8801747773315 | samimreza2111@gmail.com | samim-reza.github.io

GitHub | LinkedIn



## EDUCATION

### Green University of Bangladesh

*B.Sc. in Computer Science and Engineering*

2022 – 2025

CGPA: 3.80/4.00

### Government Bangabandhu College

*Higher Secondary Certificate*

2017 – 2019

GPA: 4.75/5.00

### S.M. Model Govt. High School

*Secondary School Certificate*

2012 – 2017

GPA: 5.00/5.00

## WORK EXPERIENCE

### Machine Learning Engineer — Brandifies

Dec 2025 – Present

- Built end-to-end ML pipelines, Fine-tuned LLMs and implemented RAG-based chatbot systems.
- Developed predictive models using XGBoost and similar boosting algorithms.

### Robotics Engineer Intern — RoboTech Valley

Jul – Sept 2025

- Designed robotics modules using embedded systems and ROS for autonomous navigation.
- Worked with Raspberry Pi, remote terminal access, and node communication.

### Teaching Assistant — Green University of Bangladesh

May – Aug 2025

- Conducted demo classes on C Programming and Data Structures.
- Evaluated exam scripts and prepared grade sheets.

### Programming Trainer — Green University of Bangladesh

Jan – Dec 2025

- Trained students for competitive programming through weekly sessions.
- Conducted classes on C, C++, Data Structures, and STL.

### Mentor — University Mentorship Program

Jun 2023 – Aug 2025

- Mentored first-year students in programming fundamentals and professional tools.
- Conducted weekly mentoring sessions.

## RESEARCH EXPERIENCE (ONGOING, 2 SUBMITTED)

### Denoising Diffusion Probabilistic Models

- Integrating edge detection for enhanced image generation.
- Working on image desmoking and restoration in adversarial scenarios.

### Federated Learning

- Adaptive client selection algorithms for optimal convergence.
- Focused on energy efficiency and security.

### Human-in-the-Loop ML

- Survey on HITL approaches in modern ML systems.

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Java, JavaScript, PHP

**AI/ML:** PyTorch, TensorFlow, Scikit-learn, LangChain, VectorDB, RAG, LLM, Roboflow

**Web/Backend:** Django, FastAPI, REST APIs, PostgreSQL, MySQL, PWA

**Tools & Platforms:** ROS, Git, Docker, LaTeX, Linux, Render, Azure, Arduino, Raspberry Pi, Jetson, ESP32

## FEATURED PROJECTS

---

- Personal AI Assistant — LLM, RAG, LangChain** Live | GitHub  
– Built an LLM-powered chatbot with RAG implementation for personalized assistance.
- KobitaLM — Bengali Poetry Generator** Model | GitHub  
– Fine-tuned Gemma 2 9B with LoRA and RAG for Bengali poetry generation.
- Union Ledger — Django, PostgreSQL, Render** Live | GitHub  
– Built a collaborative financial tracking web app with consensus-based voting for deposits, loans, repayments, and investments.
- WeCare Medical Service App — FastAPI, Ollama/Qwen3-VL, PWA, MySQL** GitHub  
– Developed an offline-first PWA for rural healthcare with AI-assisted triage capabilities.
- Cirrhosis Outcome Prediction — XGBoost, LightGBM, CatBoost** BAIC 2025  
– Built an ensemble ML classifier with 55 engineered features, achieving 14.5% accuracy gain.
- Mars Rover — ROS, Embedded Systems** GitHub  
– Developed a ROS-based autonomous navigation system for Mars rover simulation.
- Animal Introductory System — Java, MySQL** GitHub  
– Built a Java GUI application with MySQL database for animal information management.

## COMPETITIVE PROGRAMMING

---

- **Champion:** Intra Dept Programming Contest 2024.
- **3x ICPC Regionalist:** 2022, 2023, 2024.
- Around **1000** problems solved across online judges.
- Kaggle Competitor: Active in ML competitions.
- **Problem Setter:** Junior Programming Contests.
- Participated: **NCPC 2023, IUPC 2023/2024.**

## ACHIEVEMENTS

---

- **Vice Chancellor's and Dean's Award** for Academic Excellence.
- **Champion**, Inter Dept Programming Contest 2024.
- Secured **8th** position in BUBT Artificial Intelligence Conquest (Kaggle).
- National IT Quiz Competition **Medalist**.

## REFERENCES

---

### Prof. Dr. Md. Saiful Azad

Dean, Faculty of Science & Engineering  
Green University of Bangladesh  
Email: dean@fse.green.edu.bd  
Mobile: +880 1747-230418

### Ms. Umme Ruman

Assistant Professor  
Green University of Bangladesh  
Email: ruman@cse.green.edu.bd  
Mobile: +880 1799-089667